

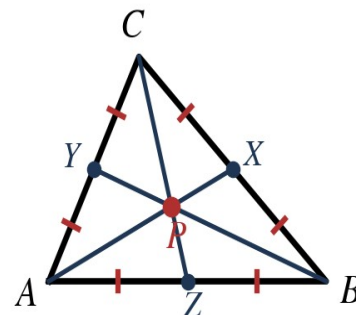
Speak Like A Mathematician:

median of a triangle:

centroid:

altitude of a triangle:

orthocenter:



The Centroid Theorem

The centroid of a triangle is _____ of the distance from each vertex to the midpoint of the opposite side.

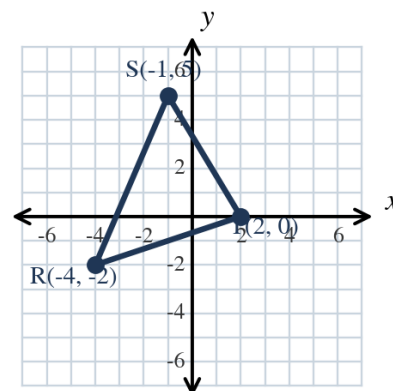
So the centroid splits every median in a _____ : _____ ratio — the long piece touches the _____.

Where the orthocenter lands: acute → _____ · right → at the _____ vertex · obtuse → _____ the triangle.

Example 1 - Find a Centroid in the Coordinate Plane

Find the coordinates of the centroid of $\triangle RST$ with $R(-4, -2)$, $S(-1, 5)$, and $T(2, 0)$.

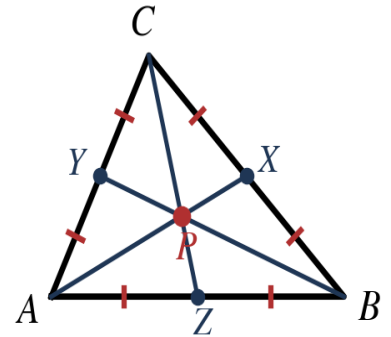
- Average the three x-coordinates.
- Average the three y-coordinates.
- State the centroid.



Example 2 - Use the Centroid Theorem

In the diagram, $\overline{AX}$, $\overline{BY}$, and $\overline{CZ}$ are medians of $\triangle ABC$ and they meet at P .

- a) $AX = 24$. Find AP and PX .
- b) $BP = 14$. Find BY and PY .
- c) $PZ = 9$. Find CZ and CP .
- d) Which piece of a median is always longer, and by how much?



Example 3 - Median or Altitude?

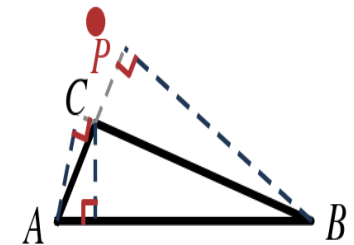
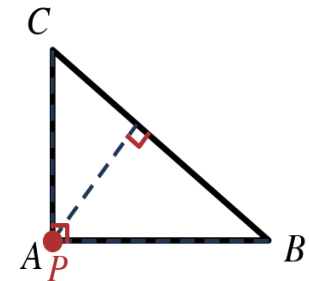
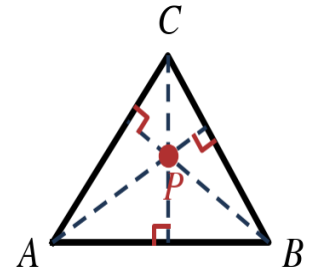
Decide whether each segment is a median, an altitude, both, or neither.

- a) A segment from a vertex to the midpoint of the opposite side.
- b) A segment from a vertex perpendicular to the opposite side.
- c) In an isosceles triangle, the segment from the vertex angle to the midpoint of the base.
- d) In a scalene triangle, can a median ever be an altitude?

Example 4 - Where Is the Orthocenter?

Each triangle at the right shows its three altitudes (dashed) meeting at P.

- Describe where P lands in an acute triangle.
- Describe where P lands in a right triangle.
- Describe where P lands in an obtuse triangle.



Example 5 - Putting the Four Centers Together

Name the point of concurrency described in each statement.

- the three perpendicular bisectors
- the three angle bisectors
- the three medians
- the three altitudes
- Which two are always inside the triangle?